Federated Learning for Activity Recognition and Face Mask Detection on Edge AI Infrastructure

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***Abstract*—Federated Learning (FL) is a decentralized machine learning approach that facilitates model training across distributed edge devices while ensuring data privacy. This study investigates the application of FL within Edge AI contexts, where edge devices with limited computational resources collaboratively train models using local data, without relying on centralized data aggregation. We demonstrate two representative use cases: activity recognition and face mask detection. The experiments are conducted using the Flower FL framework, which enables orchestration and coordination across heterogeneous edge devices. The results confirm the practicality of deploying FL on resource-constrained devices and expose key challenges, including data and device heterogeneity, communication latency, and limited processing capabilities.**

Keywords—federated learning, edge AI, activity recognition, face detection

I. Introduction

The exponential proliferation of connected devices and the growing volume of data generated at the network edge have driven the development of Edge Artificial Intelligence (Edge AI) systems. These systems enable low-latency, real-time decision-making by processing data locally on edge devices such as smartphones, sensors, and embedded platforms. However, the conventional approach of aggregating data centrally for machine learning model training introduces significant challenges, particularly concerning user privacy, bandwidth consumption, and energy efficiency. To address these challenges, Federated Learning (FL) has emerged as a decentralized machine learning paradigm that enables collaborative model training across distributed clients while retaining data locally. In this architecture, edge devices received the shared model from a central server, compute model updates using their private data, and only the learned parameters are transmitted to the server for aggregation, ensuring that raw data never leaves the local device. This approach enhances data privacy and security, minimizes communication overhead, and facilitates compliance with data protection regulations. The decentralized nature of FL makes it particularly well-suited for Edge AI deployments, where data is often sensitive, computational resources are limited, and network connectivity is intermittent or costly.

Despite its promising benefits, the application of FL in Edge AI environments introduces several technical and practical challenges. These include heterogeneity of data across clients, system heterogeneity in terms of computing capabilities and energy constraints, communication inefficiencies, and trust and security vulnerabilities such as adversarial attacks and model poisoning. Furthermore, the lack of standardized FL benchmarks and frameworks for embedded platforms poses a barrier to broader adoption and experimentation.

To advance research and practical understanding of FL in edge environments, this paper focuses on two representative use cases: Human Activity Recognition and Face Mask Detection. These applications utilize wearable sensor data collected from low-power edge devices such as the Arduino Nano 33 BLE Sense, in conjunction with the Raspberry Pi 4 for model training and FL coordination. Both scenarios are implemented and evaluated using the Flower FL framework (https://flower.ai/), which enables orchestration across heterogeneous devices and supports scalable deployment in real-world Edge AI settings. The primary objective is to assess the feasibility, performance trade-offs, and operational limitations of deploying FL on resource-constrained embedded platforms.

The main contributions of this paper are:

- An overview of the principles and challenges of FL in the context of edge computing.
- A practical implementation of FL-based activity recognition and face mask detection using widely accessible edge hardware.
- An evaluation of system performance, model accuracy, and communication costs using the Flower framework across heterogeneous clients.

The remainder of this paper is organized as follows: Section 2 reviews related work on FL and its applications in Edge AI. Section 3 details the use case scenarios, presents system architecture and experimental setup. Section 4 presents evaluation results and discussion. Section 5 concludes the paper and outlines future research directions.

II. Related Work

FL has emerged as a promising approach to decentralized machine learning, particularly relevant for edge computing scenarios where data privacy, communication efficiency, and system heterogeneity are critical. The foundational work by Li et al. [1] outlines the core challenges in FL, categorizing them into statistical heterogeneity, system heterogeneity, and privacy concerns. The authors also discuss algorithmic strategies to address these issues, such as FedAvg and personalization techniques, laying the groundwork for much of the subsequent research in the field. Treleaven et al. [2] present a broad overview of FL's significance as a privacy-

This work has been supported by the Ministry of Science, Technological Development and Innovation of the Republic of Serbia [Grant Number: 451-03-137/2025-03/ 200102].

2025 IEEE 34th International Conference on Microelectronics (MIEL) | 979-8-3315-1418-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/MIEL66332.2025.11261188

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preserving data analysis paradigm. They emphasize FL's potential to reshape sectors such as healthcare and finance by enabling secure, distributed AI systems without requiring raw data sharing. Baresi et al. [3] further explore the open research challenges in FL, including model poisoning, inference attacks, data scarcity, and fairness. They propose a research agenda centered on cross-device and cross-silo FL use cases and emphasize the necessity of standardized benchmarks and robust evaluation protocols to ensure progress and comparability across studies. A comprehensive taxonomy of FL strategies, tools, and applications is provided by Yurdem et al. [4], who classify approaches based on aggregation strategies, client selection techniques, and communication protocols. The paper surveys a wide range of FL use cases, from smart cities to autonomous systems, and identifies key open questions, such as model interpretability and resource-aware learning at the edge.

Trustworthiness is a growing concern in FL systems, and Tariq et al. [5] address this comprehensively. Their work offers a layered architecture for trustworthy FL, discussing technical aspects such as secure aggregation, differential privacy, and trusted execution environments. They also examine socio-technical concerns, including incentive mechanisms, explainability, and governance, advocating for a holistic approach to FL system design. From a tooling perspective, Beutel et al. [6] introduce Flower, an open-source FL framework designed to facilitate experimentation and deployment of FL systems. Flower supports modularity, cross-device training, and integration with popular ML libraries, making it a practical enabler for real-world FL applications and a valuable platform for academic research.

Collectively, these studies demonstrate the maturity and multidimensionality of FL as a field, while also pointing to significant unresolved challenges, especially in the context of Edge AI environments. Future work must continue to address the trade-offs between performance, privacy, and scalability to realize the full potential of FL in real-world, resource-constrained edge settings.

III. FL Demonstration Scenarios and Setup

To illustrate the practical applicability and advantages of FL in Edge AI environments, we designed and implemented two demonstration scenarios: human activity recognition and face mask detection. Both scenarios showcase how FL can enable privacy-preserving, decentralized training across heterogeneous edge devices using the Flower FL framework.

A. FL steps

The Flower framework structures FL into five key steps. First, the global model is initialized on the server using suitable machine/deep learning algorithm, with parameters defined by the dataset's labels and features. The server then distributes the initial model to all clients, either simulated or real edge devices, ensuring synchronized training. Each client performs one or more epochs of local training using the appropriate optimizer and regularization, updating the model based on its own data subset. After training, clients send their updated parameters back to the server. These updates reflect the local data variability and may be complete models or gradients. The server aggregates these using specified Federated Optimization algorithms, such as FedSGD and FedAvg, computing weighted averages based on the data volume each client processed. This step ensures that clients with more data influence the global model proportionally. The result is an updated global model that incorporates diverse, privacy-preserved local datasets. This five-step loop repeats across training rounds to progressively improve the model collaboratively.

B. Human Activity Recognition

The first use case focuses on human activity recognition and is explored through two complementary experiments.

In the first demonstration, we simulate FL training using the Human Activity Recognition Trondheim (HARTH) dataset (https://archive.ics.uci.edu/dataset/779/harth). HARTH is anannotated dataset collected from 22 subjects wearing dual 3-axial accelerometers in free-living settings over approximately two hours. To simulate a federated environment, we deployed a Flower FL server and four clients on a laptop. Each simulated client received a different subset of the HARTH dataset, representing data collected from different individuals and perform local training using logistic regression algorithm. The server receives updates from all participating clients and aggregates them into a single global model. This aggregation uses the FedAvg algorithm, which calculates weighted averages of the updates based on the amount of data each client used. This ensures that each client's data contributes proportionally to the global model, maintaining fairness and balance.

In the second demonstration, we transitioned from a simulated environment to real edge hardware. Arduino Nano 33 BLE Sense boards, equipped with a variety of onboard sensors (including motion, gesture, proximity, temperature, and barometric pressure), were used to capture real-time movement and environmental data. This sensor data was transmitted to Raspberry Pi 4 units via serial communication, which were responsible for executing the training process. Additionally, the collected data was reused in simulated clients to support hybrid experimentation. This setup enabled a combination of both simulated and physical edge clients, demonstrating the flexibility of federated learning (FL) across heterogeneous hardware platforms. Activity recognition models were collaboratively trained using local datasets from multiple Arduinos, on clients deployed on Raspberry Pi 4 devices and simulated on the laptop, illustrating the practical applicability of FL for various resource-constrained edge computers.

C. Face Mask Detection

The second scenario demonstrates face mask detection, which involves classifying images to determine whether a person is wearing a face mask or not. This application is particularly relevant in public health and surveillance contexts, where edge processing can help ensure privacy compliance and reduce cloud dependency.

For this demonstration, we used the Face Mask Detection dataset available on Kaggle (https://www.kaggle.com/datasets/belsonraja/face-mask-data set-with-and-without-mask), consisting of labeled images of masked and unmasked faces. Similar to the previous scenario, we initiated FL clients in both simulated environments on a laptop and on real Raspberry Pi 4 devices. The model is initialized on the server with specific settings (160×160 input size, softmax activation, two output classes) and then distributed to clients. Each client performs training on different subsets of the dataset using MobileNetV2 model to simulate a decentralized and privacy-aware learning environment. After training, clients send updated model

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parameters back to the server for aggregation using the FedAvg algorithm. This process repeats across multiple rounds (typically ten) to achieve faster convergence. A key enhancement in this setup is the use of Edge Impulse [7], which facilitates model deployment and testing on real edge devices as clients, such as Raspberry Pi, Arduino Nano and smartphones. After FL concludes, the global model is uploaded to the Edge Impulse platform (https://edgeimpulse.com/) via API and then downloaded by clients using login credentials. This setup enables seamless transfer and deployment of the trained model to physical edge devices, demonstrating a practical application of FL in real-world environments. We deployed the trained models on a Raspberry Pi 4 connected to an Arduino Nano BLE Sense with a camera module. The system was able to perform inference on captured images to determine mask usage, demonstrating the feasibility of deploying FL-trained models for live edge AI tasks.

IV. Experimental Evaluations

This section summarizes the results of a FL experiments for two use case scenarios.

A. FL Performance on Activity Recognition

The FL for activity recognition executed over 20 training rounds using logistic regression and the FedAvg aggregation algorithm. The model was trained once using a dataset split into 80% for training and 20% for testing. A Flower server and four Flower clients were launched, simulating distributed learning across clients. The confussion matrix for recognition of ten specified activities is shown in Fig. 1.

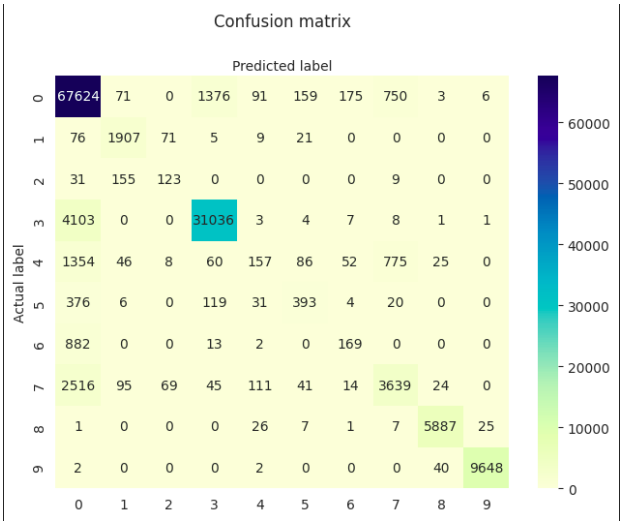


Fig. 1. Confussion matrix for FL for activity recognition.

The training process began after initializing all clients, with each client training locally and contributing updates to the global model. Accuracy significantly improved during early rounds, from 52.34% after the first round to 89.13% after the second, and 90.24% after the twentieth. Similarly, the loss dropped from 2.3025 initially to 0.3286 after round two, and 0.2869 by the final round. These results demonstrate that even a relatively simple model like logistic regression can achieve high accuracy through FL when trained over multiple rounds.

B. FL performance on face mask detection

This experiment evaluated how FL performance varies with parameters such as the number of clients, local epochs, and data batch size. The training was run with 10 communication rounds and 20 local epochs per client using a batch size of 8. As shown in the Table I and the graph (Fig. 2), increasing the number of clients generally reduced the training loss, improving overall model performance.

TABLE I. The Training Loss With tne Number of Client Changes Through Communication Rounds

Clients / Rounds	2	3	4	5
0	1.761	1.113	1.346	1.663
1	0.324	0.079	0.046	0.121
2	0.176	0.155	0.037	0.014
3	0.238	0.074	0.012	0.003
4	0.2046	0.0703	0.0056	0.003
5	0.1848	0.0547	0.0035	0.002
6	0.1656	0.0477	0.0023	0.0013
7	0.1539	0.0430	0.00164	0.00098
8	0.1480	0.0375	0.00120	0.00063
9	0.1423	0.0343	0.00091	0.00046
10	0.13804	0.0315	0.00068	0.00032

However, with more clients, aggregation on the server can lead to conflicting or lost parameter updates, which may impact model accuracy. Resource usage and communication efficiency on devices like Raspberry Pi were also considered, emphasizing the importance of balancing client count and system constraints.

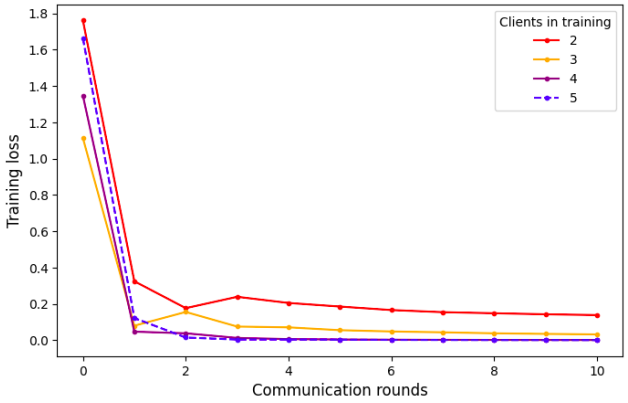


Fig. 2. The performance with tne number of client changes.

The next experiment analyzes the effect of batch size on FL performance. As shown in the graphs, smaller batch sizes (especially batch size = 4) result in lower training loss and higher accuracy. Larger batches provide more stable gradients but may not be ideal for edge devices due to memory and processing limitations. The test was conducted using two clients run on RPi and 20 local epochs. Fig. 3 illustrates the change in training accuracy across communication rounds. These results emphasize that careful tuning of batch size is crucial for optimizing performance, particularly in resource-constrained edge environments.

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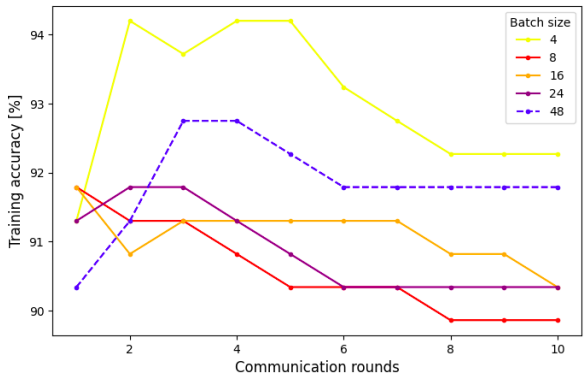


Fig. 3. The accuracy changes across communication rounds.

The next set of experiments evaluates how different numbers of local training epochs affect FL performance. It confirms that 20 local epochs yield optimal results, while increasing to 25 reduces accuracy and increases training loss. Using too few epochs can degrade model accuracy, whereas too many can cause overfitting and communication inefficiencies. The loss functions and accuracy after ten iterations of the FL process for different values of the local epochs parameter are shown in Fig. 4 and Fig. 5.

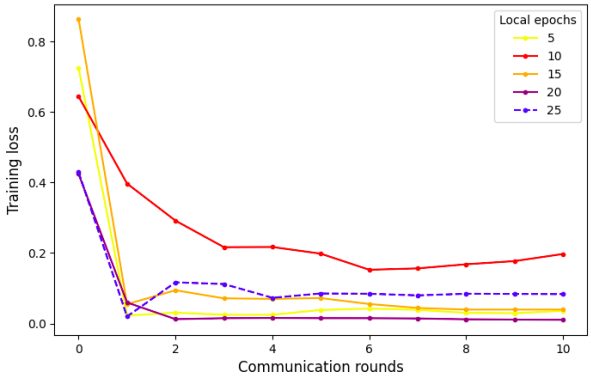


Fig. 4. Loss functions for varying number of epochs.

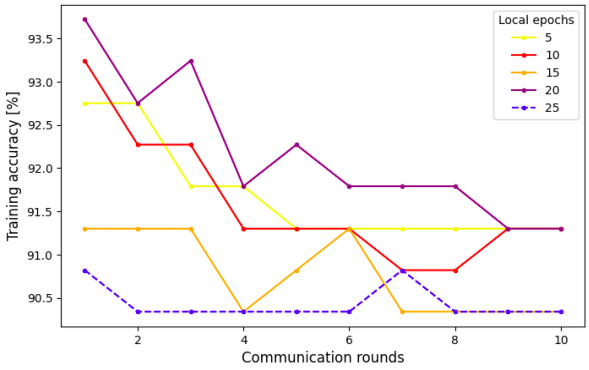


Fig. 5. FL Accuracy for varying number of epochs.

The FL experiment used two Raspberry Pi 4 devices with 20 local epochs, a batch size of 8 and 10 FL rounds to monitor memory usage, training time, and communication efficiency. Memory was tracked using the free -m command at multiple stages, showing an increase in used memory but staying within manageable levels. The memory usage remained stable between the fifth and ninth rounds, indicating efficient resource handling. The entire training process for each client lasted 12 minutes and 43 seconds in average. Communication efficiency was assessed by measuring the size of the updated model parameters sent to the server after each round, which was only 2.16 KB. This confirms that the communication overhead in the FL setup was minimal.

After training, the model was uploaded to Edge Impulse, which supports diverse edge devices (Arduio Nano, Raspberry Pi 4, smartphone,…), various model optimisation and provides detailed performance metrics. On the Raspberry Pi client device, the final model showed efficient execution with 90 ms processing time, 4.8 MB RAM usage, and 9.1 MB flash memory footprint. Model testing on Raspberry Pi devices experiences latency because image frames are first captured by an Arduino camera and transmitted over the serial port. After receiving the image, the RPi converts it into the required format before applying the trained model. This multi-step process introduces a delay but allows for real-time edge inference. The experimental output shown in confirmed high model accuracy, correctly identifying a masked face with 99.998% confidence. The output of the model applied to the image captured on cammera and executed on RPi 4 is shown in Fig. 6.

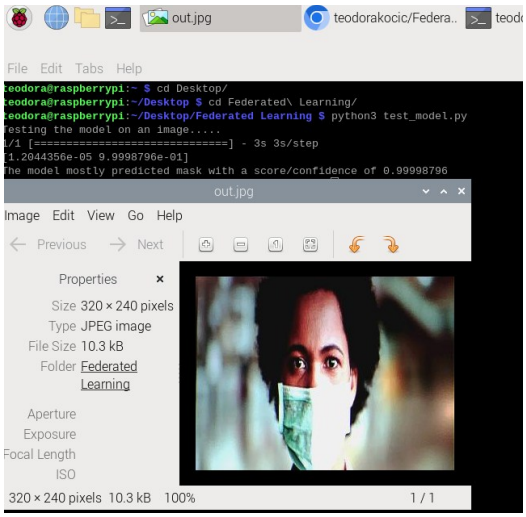


Fig. 6. Final model accuracy applied to the captured image on RPi 4.

This demonstrates the system's capability for precise detection even under resource-constrained conditions.

The source code for activity recognition (https://github.com/teodorakocic/FL_Flower) and face mask detection (https://github.com/teodorakocic/Federated_RPI) scenarios, along with data used in the experiments is available at GitHub and the results of models execution in tabular (csv) format for all use case scenarios.

V. Conclusion

This paper demonstrated the practical application of FL in edge AI scenarios through two representative use cases: activity recognition and face mask detection. By leveraging the Flower FL framework and devices such as Raspberry Pi 4 and Arduino Nano 33 BLE Sense, we showed how decentralized training can preserve data privacy while enabling model accuracy comparable to centralized approaches.

Our experiments confirmed that parameters such as the number of clients, batch size, and local epochs significantly impact the performance, efficiency, and convergence of FL models. Resource usage measurements on edge devices confirmed that FL is feasible even in constrained

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environments, with memory and flash usage staying within acceptable limits. Additionally, real-time image classification using a camera-connected Arduino highlighted the system's ability to perform accurate predictions under live edge deployment conditions.

For future research, it would be valuable to explore more advanced model architectures optimized for low-power devices, investigate communication-efficient strategies such as model quantization or sparsification, and study the effects of heterogeneous data distributions across clients. Further integration with platforms like Edge Impulse and extended evaluation on larger networks of real-world devices could enable robust, scalable, and privacy-preserving AI at the edge. Moreover, applying FL to more sensitive and dynamic domains such as healthcare or smart homes could push the boundaries of what is possible with trustworthy decentralized AI systems.

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